BEING IND-QUASI-AFFINE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19654-19657 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19659-19678, the complete author proof (18 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Short but self-contained; both directions are written.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Base-change invariance of ind-quasi-affineness and the alternative quasi-compact-open characterization (same chapter), plus the Descent chapter fpqc descent theorem for quasi-affine morphisms.

Source reference. Original lemma 02L7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19678–19678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AP7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19635–19637. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0F1U: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19538–19554. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property of being ind-quasi-affine is fpqc local on the base.*

Proof. The stability of ind-quasi-affineness under base change supplied by Lemma 0AP7 gives one direction. For the other, let $f : X \to Y$ be a morphism of schemes and let $\{g_i : Y_i \to Y\}$ be an fpqc covering such that the base change $f_i : X_i \to Y_i$ is ind-quasi-affine for all i. We need to show f is ind-quasi-affine.

By Lemma 0F1U, we may work Zariski locally on Y, so we assume that Y is affine. Then we use stability under base change ensured by Lemma 0AP7 to refine the cover and assume that it is given by a single affine, faithfully flat morphism $g : Y' \to Y$. For any quasi-compact open $U \subset X$, its Y'-base change $U \times_Y Y' \subset X \times_Y Y'$ is also quasi-compact. It remains to observe that, by Descent, Lemma 02L7, the map $U \to Y$ is quasi-affine if and only if so is $U \times_Y Y' \to Y'$. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.